Discourse-JEPA: Latent World Models over Reasoning Steps

Method and five experimental cycles — fully reproducible from these slides

TextJEPA project

July 9, 2026 (repo: `TextJEPA/`, runs: `runs/disc_*`)

Idea in one slide

- Treat a **reasoning trace** as trajectories of a world model: the *state* s_t is a compressed discourse state ("what has been established"), the *action* a_t is a tiny latent code of an *intent phrase* for the next step.
- Actions state intent only (*"derive purple apples from orange jars times green pens ."*) — never the outcome ("$= 5$"). The predictor $F(s_t, a_t)$ must therefore **compute consequences in latent space**.
- Generation $=$ **search over actions**: roll candidate actions forward with F, score with an energy, execute the best in the environment, re-encode, replan (closed-loop MPC). **No decoder, no token loss anywhere** (JEPA-pure; the symbolic environment renders chosen actions via templates).
- Synthetic reasoning world $\Rightarrow$ exact ground truth for **linear probes** and **objective planning success**.

The story in five acts (red thread)

1. **A latent world model over reasoning steps works** — and we can *audit* it: predicted futures of never-taken actions match symbolic execution (Results 1, 2, 9).
2. **What silently breaks**: EMA-target prediction deletes computed content ("collusion", Result 3); the latent metric is not a progress metric (Result 5).
3. **Reconstruction-free fixes**: frozen random-init embedding targets restore content (Result 4); trajectory-geometry losses fix the metric (Results 7, 8); counterfactual *ranking* fixes the energy's discrimination (Result 10).
4. **The energy can come from the geometry itself**: a value head distilled from the latent goal distance, with zero numeric supervision, beats the fully supervised one (Results 13, 16).
5. **Baselines and limits**: likelihood selection (LMs at any tested scale/granularity) clusters far below (Result 15); failures are chain-depth-limited — the scale-up frontier (Result 16).

Glossary — every term and run-name used in this deck

discourse state s_t	one vector summarizing prompt + steps $1..t$ (output of the causal state model)
predictor $F(s, a)$	MLP producing the next state from state + action code (the world model)
action a	16-d encoding of an *intent phrase* ("derive X from Y times Z") — states what to do, never the outcome
value head $V(s, s_0)$	scalar energy: estimated necessary steps remaining; the planner minimizes steps + V
value-grad	value-head gradients flow into the encoder (`value_detach=false`) so the energy shapes the representation
LDAD	Latent Displacement Action Decoding (Delta-JEPA): decode the executed action from $\Delta s = s_{t+1} - s_t$
anchor	auxiliary target: predict the *frozen random-init* encoder's embedding of the next sentence (collusion fix)
ranking (K)	margin loss ordering $V(F(s, a_i))$ over K counterfactual actions per state by their symbolic outcomes
cost ranking	same, over multi-step costs (depth + V) — calibrates search depth
hinge / monotonicity	necessary steps must decrease the latent distance to the trace-terminal state by a margin
distilled V	value head regressed on that latent goal distance (geometry target, no numeric labels)

THE recipe (sheared) and headline results (mean $\pm$ std, 3 seeds)

Claim: Five losses suffice — everything else adds $\leq$1–3% and is reported in the shear log.

	@optimal	@slack-2
random feasible policy	0.055	0.405
token LM, parameter-matched (best lr)	0.670	0.910
token LM, $3\times$ larger	0.735	0.940
core recipe (`core5`)	**0.880 $\pm$ 0.004**	**0.975 $\pm$ 0.004**
+ depth option (full stack, look-2)	0.945	1.00
symbolic oracle	1.00	1.00

core5 = latent prediction + counterfactual ranking + VICReg + frozen-anchor prediction + LDAD; the value head is trained by ranking margins alone — *no numeric value labels*. The depth option (cost ranking, look-2) needs the fuller stack (Results 12, 16); on the sheared recipe it does not pay (0.885 look-2 vs 0.900 look-1).

Value-head supervision targets compared (symbolic vs geometric)

Claim: The geometric target
(latent goal distance) with ranking beats numeric-label regression — IF the metric is shaped first.

V trained on	metric shaping	labels used	@opt	@slack2
numeric remaining-steps	—	scalar counts	0.64±.05	0.89
numeric + ranking	—	counts + order	0.91±.03	0.99
numeric + ranking + hinge	hinge	counts + order + binary	0.89±.02	0.99
latent goal distance	**none** (control)	none	0.19	0.64
latent goal distance	curvature (label-free)	none	0.30	0.72
latent goal distance	hinge	binary relevance	0.57±.05	0.89
lat. dist. + ranking	hinge	binary + order	**0.90±.03**	**0.98**

Reading: the distilled (geometric) value head is a drop-in replacement for numeric supervision once the metric it distills is goal-monotone; scalar count labels contribute nothing beyond binary relevance + ordering (0.90 vs 0.89/0.91, within seed noise).

World: iGSM-style synthetic reasoning (exact generator)

Problem sampler (rejection-sampled until $3 \leq |\mathcal{A}_q| \leq 9$; `rng = Random(f"{seed}:{index}")`):

- $n \sim \mathcal{U}\{6, \ldots, 12\}$ variables; names = distinct (adjective, noun) pairs from 20×20 vocabulary.
- Variable i: with p=0.35 (always for $i < 2$) a **leaf** $x_i = c_i,\ c_i \sim \mathcal{U}\{0..22\}$; else an **internal** $x_i = x_a \circ x_b \bmod 23$, $\circ \in \{+, -, \times\}$, parents a, b sampled from $\{0..i-1\}$.
- Query q = uniform over internal variables; $\mathcal{A}_q$ = ancestors of q (incl. q) = the necessary computation; the rest are **distractors**.

Training traces: solve by repeatedly resolving a feasible variable (parents resolved); with p=0.15 (max 2) pick a feasible *distractor* instead of a necessary one $\Rightarrow$ off-path states for the value head. Fresh problems every epoch (train seed 1, val seed 2; 100k/5k).

A verbatim sample (val problem #3)

Prompt (definitions shuffled per-sample, question last; one sentence = one chunk):

the number of purple apples equals the number of orange jars times the number of green pens .
the number of orange jars is 15 . *the number of green pens is 8 .*
(+ 7 distractor definitions) *how many purple apples are there ?*

Solution trace — step sentence (observed outcome) vs. action intent phrase (no outcome):

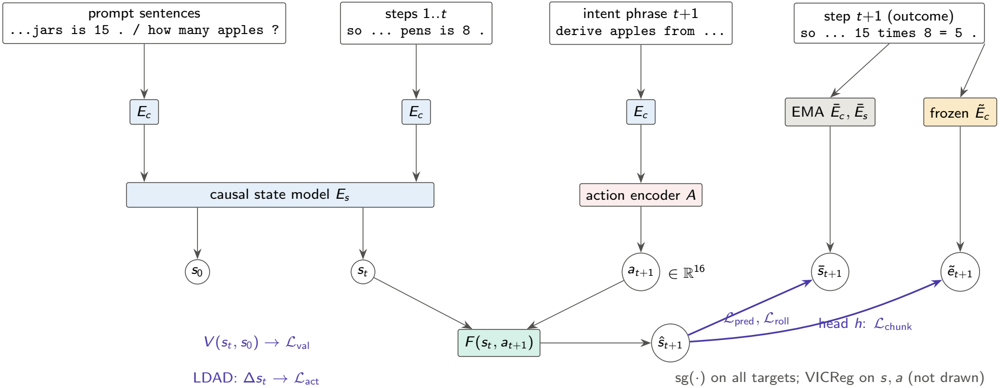
step sentence (encoder sees)	intent phrase (action encoder sees)
`so the number of green pens is 8 .`	`look up the number of green pens .`
`so the number of orange jars is 15 .`	`look up the number of orange jars .`
`so the number of purple apples is 15 times 8 = 5 .`	`derive purple apples from orange jars times green pens .`

Note $15 \times 8 = 120 \equiv 5 \pmod{23}$: outcomes are nontrivial to predict.

Architecture (all sizes exact; 8.8–9.1M params)

- **Chunk encoder** E_c: word-level vocab (441 tokens), embed 256, 2-layer pre-norm transformer (4 heads, FF 1024), masked mean-pool $\rightarrow$ one 256-d vector per sentence. Shared for prompt, steps, intents.
- **State model** E_s: 4-layer *causal* transformer (8 heads, FF 1024) over [prompt chunks | step chunks] + learned positions + 2 segment embeddings. s_0 = hidden at the question; s_t = hidden at step t.
- **Action encoder** A: MLP 256 $\rightarrow$ 256 $\rightarrow$ 16 on the intent-phrase embedding (optional FSQ, unused in main runs).
- **Predictor** F: residual MLP, $\hat{s}_{t+1} = s_t + \mathrm{MLP}([\mathrm{LN}(s_t); a_t])$, 2 hidden layers of 1024 (GELU).
- **Hierarchy**: CLS-transformer over K=3 action codes $\rightarrow$ 8-d macro action m_t; $F_{\mathrm{hi}}(s_t, m_t)$ predicts $\bar{s}_{t+3}$ (same latent space, HWM-style).
- **LDAD** (Delta-JEPA): MLP on $\Delta s = s_{t+1} - s_t \rightarrow$ op logits (4) + regression to the EMA intent embedding.
- **Value head** V: MLP on $[\mathrm{LN}([s_t; s_0])] \rightarrow$ scalar = predicted remaining necessary steps.
- **Teachers**: EMA copies of E_c, E_s (momentum 0.99 $\rightarrow$ 0.999 linear); **frozen anchor** $\tilde{E}_c$ = copy of E_c at random init, never updated.

Training pipeline



Objectives (exact losses and weights)

Let $d(x, y) = \mathrm{smoothL1}(\mathrm{LN}(x), \mathrm{LN}(y))$ (per-dim mean; LN = parameter-free layer norm), masks over valid steps.

$$
\begin{array}{rll}
\mathcal{L}_{\text{pred}} = & d\big(F(s_t, a_t),\, \mathrm{sg}\, \bar{s}_{t+1}\big) & \text{weight } 1.0 \\
\mathcal{L}_{\text{roll}} = & d\big(F^{(t)}(s_0, a_{1:t}),\, \mathrm{sg}\, \bar{s}_t\big) \text{ (open loop, all } t\text{)} & 1.0 \\
\mathcal{L}_{\text{hi}} = & d\big(F_{\mathrm{hi}}(s_t, m_t),\, \mathrm{sg}\, \bar{s}_{t+3}\big) & 0.5 \\
\mathcal{L}_{\text{vic}} = & \sum_j \max(0, 1 - \mathrm{std}(s^{(j)})) + 0.04\, \|\mathrm{offdiag}(\mathrm{Cov}(s))\|_F^2 / D + 0.1\, \mathrm{var}(a) & 1.0 \\
\mathcal{L}_{\text{act}} = & \mathrm{CE}\big(\mathrm{op} \mid \Delta s_t\big) + d\big(g(\Delta s_t),\, \mathrm{sg}\, \bar{E}_c(u_t)\big) \quad \text{(Delta-JEPA)} & 2.0 \\
\mathcal{L}_{\text{val}} = & \mathrm{smoothL1}\big(V(s_t, s_0),\, \#\text{unresolved necessary}\big) & 1.0 \\
\mathcal{L}_{\text{chunk}} = & d\big(h(\hat{s}_{t+1}),\, \mathrm{sg}\, \tilde{E}_c(y_{t+1})\big) + 0.5\, d\big(h(\text{rollout}_t),\, \mathrm{sg}\, \tilde{E}_c(y_t)\big) & 0 \;/\; 2.0 \\
\mathcal{L}_{\text{curv}} = & 1 - \cos(v_t,\, v_{t+1}), \quad v_t = s_{t+1} - s_t \text{ (straightening, arXiv:2603.12231)} & 0 \;/\; .02\text{–}.1 \\
\mathcal{L}_{\text{mono}} = & \mathbb{1}_{\text{nec}}\, [\Delta d_t^{\text{goal}} + m]_+ + \tfrac{1}{2}(1 - \mathbb{1}_{\text{nec}})\, [-\Delta d_t^{\text{goal}}]_+ & 0 \;/\; 1\text{–}2
\end{array}
$$

$\mathcal{L}_{\text{chunk}}$: weight 0 in `base`; 2.0 in `chunkpred`/`combo` (**frozen** anchor $\tilde{E}_c$). Value head input detached except `valgrad`/`combo`. d_t^{goal} = LN-L1 distance of s_t to the trace-terminal EMA state; margin $m \in \{.02, .05\}$.

Training protocol

- AdamW, lr 3×10^{-4}, weight decay 0.05 (none for dim < 2 params), $\beta = (0.9, 0.95)$; cosine schedule to floor 0.05, 1000-step warmup; grad clip 5.0.
- Batch 128 problems; 20 epochs $\times$ 781 steps; **fresh data every epoch** (no memorization confound). EMA momentum 0.99 $\rightarrow$ 0.999 linear over training.
- Hardware: one V100-32GB, fp32, $\sim$40 min/run.
- In-training eval each epoch: all losses + collapse diagnostics (per-dim std, RankMe effective rank) + closed-form ridge probes.
- Code: `scripts/train.py +experiment=...`; every run's exact config is stored in its checkpoint.

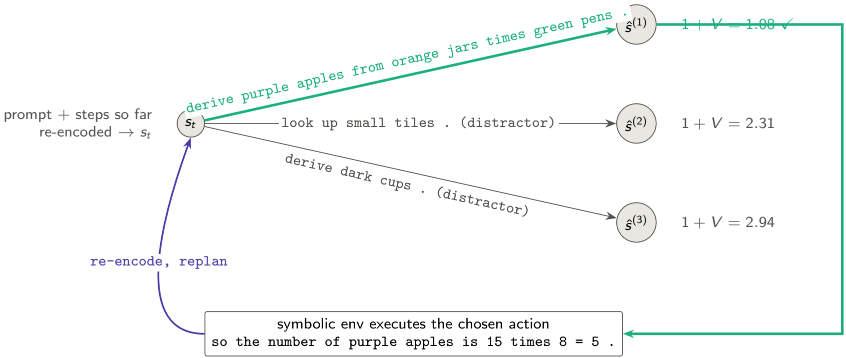
Planning: closed-loop MPC by exhaustive latent search

Interface knowledge (legitimate): feasible actions = unresolved variables whose stated dependencies are resolved — readable from the prompt. **Consequence knowledge**: latent only.

1. Encode prompt + steps so far $\rightarrow s$ (and s_0 once).
2. Enumerate feasible action *sequences* to depth L (lookahead; cap 64), encode their intent phrases $\rightarrow a_i$.
3. Roll out $\hat{s} \leftarrow F(\hat{s}, a_i)$ along each sequence.
4. Score: $\text{cost} = \text{steps taken} + V(\hat{s}_{\text{end}}, s_0)$ (estimated total steps).
5. Execute the first action of the argmin in the symbolic environment $\rightarrow$ real outcome sentence appended; go to 1.

Success = query resolved within the **strict optimal budget** $|\mathcal{A}_q|$ (slack 0: any distractor pick is fatal). 200 val episodes. Discrete enumeration replaces CEM; a sampled action prior would need CEM/MPPI.

Planning: search graph (one replanning round, lookahead 1)



V values are predicted *remaining necessary steps* of the predicted next latent; lookahead 2 scores 2-step latent rollouts.

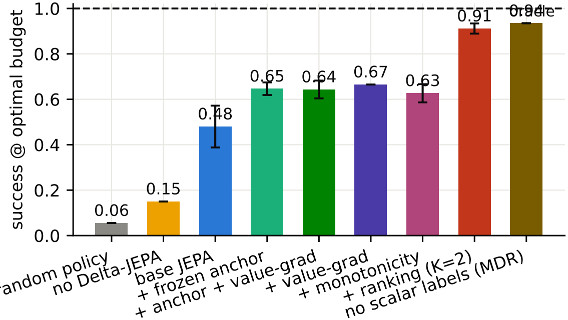
Experiment grid (each = one training run, 20 epochs)

run	$\mathcal{L}_{\text{chunk}}$ anchor	value grad $\rightarrow$ encoder	other
`base`	—	no	reference
`no_delta`	—	no	$\mathcal{L}_{\text{act}}$ weight 0
`chunkpred`	frozen, w=2.0	no	—
`valgrad`	—	**yes**	—
`combo`	frozen, w=2.0	**yes**	—
`value_only`	—	**yes**	*all* JEPA weights 0
`frozen_enc`	—	no	encoders frozen at init
`straight_{lo,mid,---}`	frozen, w=2.0	**yes**	$+\,\mathcal{L}_{\text{curv}}$, $\lambda \in \{.02, .05, .1\}$
`mono` / `_hi`	frozen, w=2.0	**yes**	$+\,\mathcal{L}_{\text{mono}}$, w1 m.02 / w2 m.05
`geo_lo` / `_geo`	frozen, w=2.0	**yes**	both regularizers
`mono_novalue`	frozen, w=2.0	no	$+\,\mathcal{L}_{\text{mono}}$, *no value head*
`rank_{k2,k4}(_nodelta)`	frozen, w=2.0	**yes**	+ ranking over K counterfactual actions
`champion`	frozen, w=2.0	**yes**	ranking K=2 + $\mathcal{L}_{\text{mono}}$ w2
`{shuffled, no_distract, size*}`	frozen, w=2.0	**yes**	grounding controls (Results 9, 11)

Baselines at plan time: **random feasible policy**, **symbolic oracle** (=100%), and **oracle goal-distance energy** (LeWM-style, uses the encoded solved state — diagnostic only).

Result 1 — training objectives determine plannability

Claim: The anchor lifts the base model decisively; value-grad and the hinge sit within seed noise of it; counterfactual ranking is the step change. Bars: mean ± std over 3 seeds (thin caps).



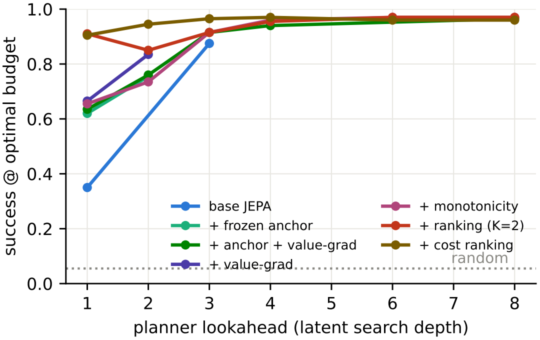
Success within the *strict optimal budget*, greedy lookahead 1, 200 episodes, mean necessary steps 4.07.

With slack 2 (mean of 3 seeds): base 0.83, +anchor 0.89, +hinge 0.91, +ranking / MDR **0.98–0.99** (random 0.41).

Distractor pick rate: random 0.43 → base 0.28 → valgrad 0.14.

Result 2 — search depth is a planning-time compute knob

Claim: Success grows with search depth up to a shared ≈0.97 ceiling: rollouts of F and the energy stay trustworthy over many imagined steps, and depth even rescues weak energies.



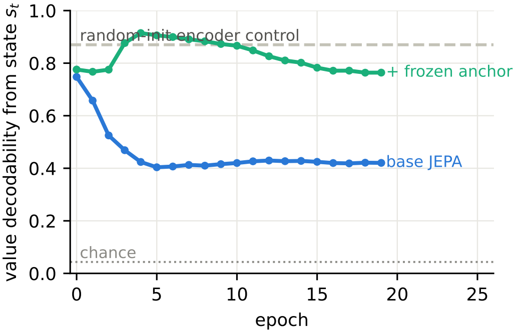
Lookahead d = feasible d-step sequences (capped), scored at the imagined end state; no retraining. Depth curves are single-seed; the depth-1 anchors carry ±0.03–0.05 seed bands (leaderboard slide).

All recipes converge to ≈0.96–0.97 by depth 4 — the residual is the deep-chain failure mass (Result on failures), not search.

Base JEPA: 0.35 → 0.87 (depth 3); `mono_hi` 0.655 → 0.97 (depth 6). The `rank_k2` dip at depth 2 is the calibration anomaly; cost ranking (brown) removes it and dominates at every depth.

Result 3 — encoder–predictor collusion, measured

Claim: Pure state-prediction JEPA *removes* outcome information that was linearly present at initialization; a frozen outcome anchor prevents it.



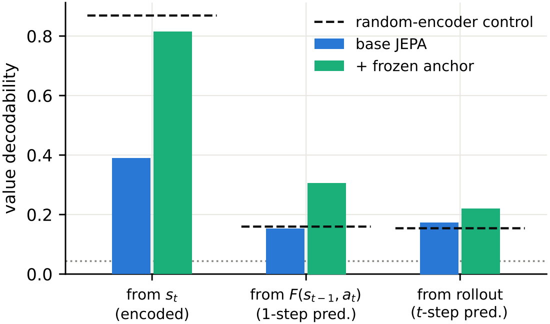
Probe: linear decodability of the just-computed value (23 classes) from s_t, val split.

Both loss sides are learned; EMA targets *trail the online encoder*, so deleting a hard-to-predict feature lowers the loss — stop-grad does not pin content.

Consequence in base: answer decodable from the terminal state at only 0.26 (random-init control: 0.65).

Result 4 — latent arithmetic under the frozen anchor

Claim: With $\mathcal{L}_{\text{chunk}}$, the predictor computes outcomes it never observes: value decodability from *predicted* latents doubles the random-encoder control.



Dashed = random-init encoder control (surface memory, no computation); dotted = chance 1/23.

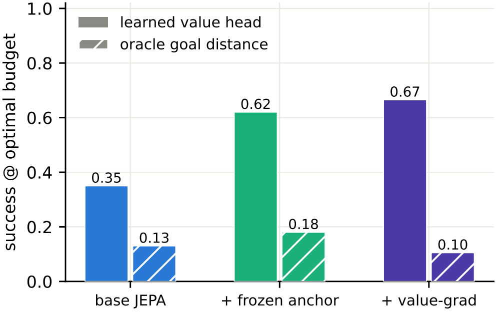
From $F(s_{t-1}, a_t)$: base 0.15 (= control) → anchor **0.31**.

Answer from terminal state: 0.26 → **0.98**.

Full-trace open-loop rollout: 0.22 (error compounds; matches Result 2's need for re-encoding).

Result 5 — the goal energy must be learned (LeWM test)

Claim: Raw latent distance to the oracle goal state plans barely above random; the trained value head is the energy that works.



Oracle goal = encoded terminal state of the *true* solution (planner gets it for free — upper-bound diagnostic).

Scoring $\|\mathrm{LN}(\hat{s}) - \mathrm{LN}(s_{\text{goal}})\|_1$: base 0.13, anchor 0.18, valgrad 0.105 (distractor rates ≈ random).

⇒ *unregularized* JEPA geometry is not a distance-to-goal metric. Result 7 shows temporal straightening fixes exactly this.

Result 6 — Delta-JEPA ablation; stability

Claim: Displacement-level action decoding is load-bearing for planning (replicates Delta-JEPA, arXiv:2606.31232, in language).

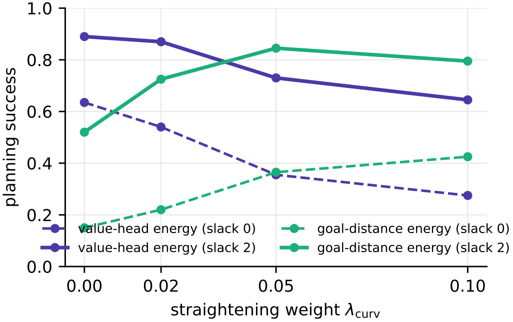
	success@opt	success@slack2	distractor rate	op from Δs
`base`	**0.35**	**0.75**	0.28	1.00
`no_delta`	0.15	0.51	0.40	0.98*

*op stays decodable (it correlates with surface features) but transitions lose the *action-discriminative* structure the planner needs.

Stability (all runs): per-dim state std ≈ 1.03, RankMe effective rank 220–243 / 256, no collapse; VICReg + EMA sufficed, discrete codes (FSQ) not needed.

Result 7 — temporal straightening makes raw geometry plannable

Claim: The curvature loss (arXiv:2603.12231) turns latent distance into a progress metric: goal-distance planning goes 0.15 → 0.43 @optimal (0.52 → 0.85 @slack 2) — but it *trades off* against the value head.



Dose-response over $\lambda_{\text{curv}} \in \{0, .02, .05, .1\}$ (all on the combo recipe):

	0	.02	.05	.1
velocity cos	−.25	−.01	.42	.67
corr(d^{goal}, remaining)	.73	.75	.80	.86
value probe from s_t	.89	.89	.74	.43

Geometry improves exactly as content decodability degrades; $\lambda = .05$ is the raw-geometry sweet spot.

Result 8 — two energies, one metric: the content–geometry trade-off

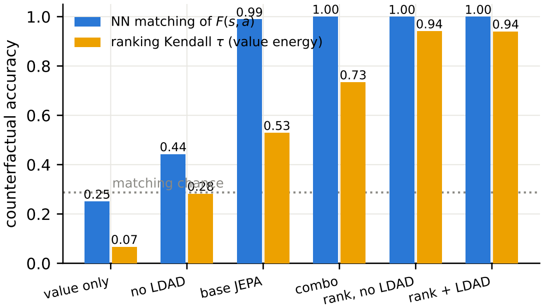
Claim: Straightening buys geometry planning, the monotonicity hinge keeps content and sets the value-planning record; geometry needs JEPA losses, not value labels.

	goal-distance energy		value-head energy	
run	@opt	@slack2	@opt	@slack2
`combo` ($\lambda = 0$)	0.15	0.52	0.635	0.89
`straight_mid` (λ=.05)	0.365	**0.845**	0.355	0.73
`mono_hi` (hinge w2 m.05)	0.255	0.68	**0.655**	**0.935**
`mono_novalue` (*no* V)	0.20	0.65	—	—
`value_only` (*no* JEPA)	0.06	—	0.075	0.43
random	0.06	0.095	0.06	0.095

Three readings. (i) One latent metric cannot currently maximize both energies — a learned *projection head* for the geometric energy is the obvious fix (tested in Result 10: it softens but does not remove the trade-off). (ii) A model with **no value head** still plans at 0.65 @slack 2 from pure geometry. (iii) `value_only`'s geometry is *exactly random*: planning-relevant structure comes from the JEPA objectives, not the value labels.

Result 9 — the predictor is counterfactually grounded (direct audit)

Claim: From each state we enumerate *all* feasible actions, predict their next-state latents, and check against symbolic execution the model never saw: $F(s, a)$ is correct per action, and LDAD / outcome anchors are what ground it.



Audit metrics (100 episodes, val): nearest-neighbor *matching* of $F(s, a_i)$ to the executed-and-encoded true next states; Kendall τ of the value energy vs. true post-action remaining steps.

Training data shows **one action per state** — grounding comes from cross-problem generalization alone.

Shuffled-action control: matching 0.30 $\approx$ chance. RSA of pairwise next-state geometry: 0.93 (base+LDAD) vs 0.48 (no LDAD).

Result 10 — counterfactual ranking: the strongest lever

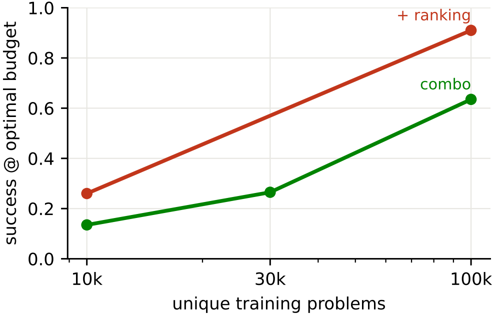
Claim: Hinge-ranking K alternative actions per state by symbolic outcome nearly closes the gap to the oracle; K=2 suffices; it can replace LDAD but not data diversity; it trades search-depth calibration for 1-step precision.

	@opt	@slack2	@opt look-2	value τ
`combo`	0.635	0.89	0.76	0.73
`rank_k2`	0.910	0.985	0.85 $\downarrow$	0.94
`rank_k4`	0.925	0.98	0.85 $\downarrow$	0.94
`rank_k4` *no LDAD*	0.875	0.965	—	0.94
`combo` *no LDAD*	0.48	0.805	—	0.69
`champion` (rank+mono)	**0.905**	**0.99**	—	**0.96**

$\downarrow$ lookahead-2 *hurts* ranking models (0.91→0.85) while helping regression values (mono_hi 0.655→0.735): the margin loss perfects 1-step ordering but distorts V's absolute scale that multi-step cost sums need. `champion` also has the best raw geometry (τ_{goal} 0.82, oracle-goal planning 0.655, no curvature loss). Seed replicates: rank_k2 0.885–0.91 @opt, 0.98–0.985 @slack2.

Result 11 — data diversity is the counterfactual fuel

Claim: Grounding comes from cross-problem coverage: shrinking unique problems collapses planning; ranking supervision sharpens but cannot substitute.



Same optimizer steps, fewer unique problems:

problems	10k	30k	100k
combo @opt	0.135	0.265	0.635
+ ranking	0.26	—	0.91
matching	0.82	1.00	1.00
value τ	0.22	0.44	0.73

Matching survives small data; *ranking quality* does not — the predictor stays roughly right, the energy loses discrimination. Within-trace variation matters too: no distractors $\Rightarrow$ 0.635$\rightarrow$0.415.

Result 12 — depth calibration: CostRanking and macro jumps

Claim: Plain ranking breaks lookahead (0.91→0.85); two independent fixes exist, and they don't stack because they repair the same defect.

	look-1	look-2	look-3 flat	look-3 F_{hi}
`rank_k2`	0.910	0.850 ↓	0.915	**0.970**
`rank_cal` (+CostRanking)	0.905	**0.945**	—	0.915
`champion_cal`	0.910	0.945	—	0.915

CostRanking: rank full MPC costs across depths — executed 2-step continuation $(2 + V(F^2))$ vs 1-step alternatives $(1 + V(F(a')))$; zero new data. **Macro jumps**: score 3-step sequences with one F_{hi} call — worse *fidelity* than F^3 (matching 0.62–0.70 vs 0.77–0.91) but stays on V's training distribution. 1000-episode: rank_cal look-2 = 0.946 vs hier 0.931 — calibration wins once you have it; macro jumps are the no-retraining workaround. Train-side hierarchy loss: neutral (`combo_nohier` 0.670/0.910).

Result 13 — the non-symbolic value head (headline)

Claim: Shape the metric with the hinge, distill it into V, add ranking: best planner with no numeric value labels anywhere.

run	energy supervision	@opt	@slack2
`champion_cal`	*numeric* + ranking	0.91 ± .01	0.995
`mono_distill_rank`	**binary only (geometric V)**	**0.90** ± **.03**	**0.98** ± **.01**
`mono_hi`	*numeric* regression	0.63 ± .05	0.91 ± .02
`mono_distill`	binary (no ranking)	0.57 ± .05	0.89
`mono_lf_distill`	*none* (all-steps hinge)	0.355	0.775
`geometric`	*none* (curvature)	0.300	0.715
`distill_v`	*none, unshaped* (control)	0.190	0.640

Rows 1 vs 2 are the direct **numeric-vs-geometric** comparison at matched search settings: the geometric value head (V regressed on the latent goal distance d^{goal}, ordered by binary-outcome ranking) matches the numeric-label champion within seed noise (0.90±.03 vs 0.91±.01). Shaping is required: unshaped distillation fails (0.19). Mean±std over 3 seeds where shown; single-seed otherwise.

Result 14 — stability: the Delta-JEPA claim does not transfer; VICReg is load-bearing

Claim: LDAD alone does not prevent collapse in language JEPA, and VICReg is not redundant even with EMA + stopgrad + frozen anchor.

run	EMA	sg	VICReg	result
`ldad_pure`	—	—	—	**collapse** (std .027, rank 80)
`ldad_pure_sg`	—	✓	—	collapse (0.11 @opt)
`combo_novic`	✓	✓	—	0.27 @opt (vs 0.635)
`combo`	✓	✓	✓	0.635 @opt

Honest negative replication of arXiv:2606.31232's stability claim in our domain (their evidence is visual control; our traces/action space differ). Also: non-residual predictor $\hat{s} = \mathrm{MLP}([\mathrm{LN}(s); a])$ performs at least as well as residual (0.710/0.920) — the residual skip is convenience, not mechanism; LDAD constrains *encoder* displacements, not predictor outputs, so it is not trivialized by either choice.

Result 15 — the baseline ladder (action-selection parity)

Claim: Every likelihood/reconstruction selector — token- or sentence-level, any tested scale — clusters at 0.47–0.74; explicit counterfactual training adds 20–30 points on top of all of them.

model (selection energy)	params	@opt	@slack2
sentence-LM (decoder CE)	7.2M	0.470	0.810
sentence-LM+latent tgt (decoder CE)	7.2M	0.510	0.800
sentence-LM+latent tgt (**latent dist.**)	7.2M	0.605	—
token LM (likelihood, best lr)	8.2M	0.670	0.910
naive JEPA (combo, value head)	9.1M	0.635	0.890
token LM	25.5M	0.735	0.940
JEPA + ranking (MDR / +cal)	9.1M	**0.935**	**0.99–1.00**

Parity protocol: every model scores the *same* feasible candidates' rendered step sentences (LM: log-likelihood; sentence-LM: decoder CE or latent distance; JEPA: latent energy), same budget. LR swept for both families (LM prefers 10^{-3}, JEPA 3×10^{-4}). Note the within-model dissociation: the sentence-LM's *latent* energy outranks its own decoder (0.605 vs 0.510) — selection is a latent-geometry problem even for reconstruction-trained models.

Result 16 — the shear log (how the recipe was found)

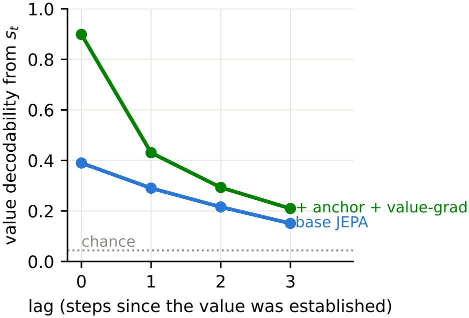
Claim: Build up from nothing; shear down from everything: five components carry the result, each further addition is $\leq$1–3% ($\approx$ seed noise).

build-up	losses	@opt	eff. rank	**sheared (from full)**	Δ@opt
ranking only	1	0.20	**24 (collapse)**	hinge	-0.02
+ latent prediction	2	0.78	101	distilled V	-0.04
+ VICReg	3	0.745	234 (healthy)	rollout supervision	$\sim$0
+ anchor + LDAD = **core5**	5	**0.88±.004**	234	hierarchy loss	$\sim$0
full stack (MDR)	9	0.90±.03	233	EMA teacher	-0.03
				scalar value labels	$\sim$0
				residual skip	$+0.02$

Each sheared component retains a specific *auxiliary* role: hinge + distilled V = the fully label-free-energy variant (0.36–0.57); EMA = +0.03 over stopgrad+VICReg (which suffice for stability — `noema` 0.605, no collapse); cost ranking = depth calibration, but only on the fuller stack (on core5: 0.885 look-2 $<$ 0.900 look-1; on the full stack: 0.945). **Failures remain chain-depth-limited**: 4.4% (3–4 steps) $\rightarrow$ 59% (7–9). Edit track mirrors the shear: `edit_core5` 0.455/0.485 = champion, no defect labels.

Probes I — what the discourse state actually is

Claim: A recency-weighted working memory over a relational scaffold: structure is objective-independent, content is objective-dependent.

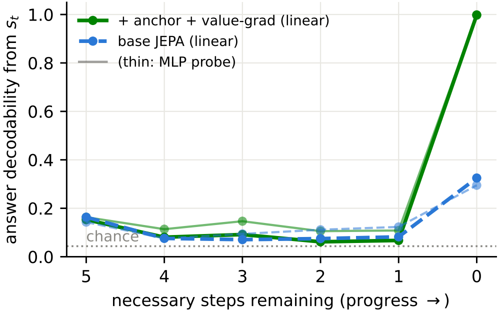


probe (combo / all recipes)	acc	maj.
resolved-set member $[s_t; \mathrm{oh}(v)]$	0.82	0.67
ancestor-of-query $[s_0; \mathrm{oh}(v)]$	0.76	0.56
var just resolved (from Δs)	0.33	0.19
query identity (from s_0)	0.19	0.16

These rows are **identical across every training recipe** (base, no-LDAD, geometry, ranking) while value decodability swings 0.39–0.90: the encoder learns the scaffold from text alone; the objectives decide what content survives. Relevance is encoded *relationally* (0.76), not as a query pointer (0.19).

Probes II — emergence, memory, circular code

Claim: No anticipatory answer computation: the answer appears exactly when its variable resolves — and only collusion-free models can hold it.



Emergence is a step function: 0.06–0.18 mid-trace (thin MLP lines: small nonlinear surplus), 1.00 the moment the query resolves; base manages only 0.33 even then (collusion).

Memory decays with lag (Probes I): 0.90 → 0.43 → 0.29 → 0.21 (chance 0.04) — a working buffer, not a symbol table.

Circular coding: ridge R^2 for $\cos(2\pi v/23)$ from s_t: 0.52 (combo) vs 0.20 (base) — the mod-p value is partly geometric, and the circle sharpens under the anchor fix.

Probe suite (champion vs. reference; random-encoder control)

probe (linear, val split)	`base`	`chunkpred`	random-enc
value of current step from s_t	0.39	**0.81**	0.87
value from $F(s_{t-1}, a_t)$ (latent arithmetic)	0.15	**0.31**	0.15
value from open-loop rollout	0.17	**0.22**	0.15
op type from Δs	1.00	1.00	0.97
step goal-relevance from Δs	0.89	**0.93**	0.89
remaining steps from s_t	0.44	**0.61**	0.41
resolved count from s_t	0.97	0.92	0.95
final answer from terminal state	0.26	**0.98**	0.62
answer from prompt alone s_0 (should be hard)	0.035	0.05	0.04
value-head MAE (remaining steps)	0.76	0.57	—

`combo` adds value-grad: MAE **0.38** with the same collusion-free probes (state value 0.89).

Scale-up preview — the hard domain and faithful iGSM

Claim: On 10–18-step problems everything collapses — the likelihood selector hardest — and the ranking/capacity/search recipe degrades most gracefully. This is the quantified frontier for the next phase.

hard domain (14–24 vars, 10–18 steps)	@opt	@slack2	look-2	look-4
reference recipe (`combo`, 9M)	0.000	0.070	0.035	0.175
minimal-labels + cost rank (`mdr_cal`, 9M)	**0.120**	**0.445**	0.105	0.150
same recipe, 35M	0.090	0.410	0.195	**0.315**
token LM (8M, parity selection)	0.010	—	—	—

Capacity only pays *through search* (35M: 0.09 greedy → 0.32 at look-4). And on **100%-faithful iGSM** (official facebookresearch generator vendored; our env drives the reference renderer; solutions pass the official checker 60/60 med, 99/100 hard): the reference recipe is near-random (0.245 vs 0.20; probes healthy — the energy fails, the pre-ranking signature), 60 epochs don't help (0.235), and **counterfactual ranking is again the fix: 0.410 @optimal / 0.735 @slack-2**, and capacity stacks on it (35M: 0.460/0.765, value probe 0.53→0.93) — the recipe's central mechanism transfers to the official domain, with the remaining gap to oracle as the next phase's frontier.

Honest limitations & what this is not

- **iGSM-style, not iGSM**: flat DAG, binary mod-23 ops, one question template, templated English — no hierarchical categories / instance-vs-abstract parameters of the original benchmark.
- **Actions are observed intents**, not variational latents: the symbolic world offers an enumerable interface. The variational prior $p(a \mid s, g)$ (for CEM/MPPI over sampled actions) is future work.
- **The best goal energy is still supervised** (remaining-steps labels); straightened raw geometry closes much of the gap (0.845 vs 0.935 @slack 2, Results 7–8) and the curvature loss is *label-free*, but the monotonicity hinge uses the same labels as the value head.
- Value labels shape only the value head unless `valgrad`; the op-CE in LDAD uses the executed action's type (JEPA-legit: actions are observed at training time).
- Text "generation" is action selection + template rendering; a learned renderer is orthogonal to the world model.

Next steps

- **Depth-calibrated ranking**: rank multi-step rollout costs, not only 1-step candidates, so lookahead helps ranking models too (Result 10's anomaly); target $\geq$0.95 @optimal with look-2.
- **Error analysis of the last 10%**: what do the `champion` failures share (chain depth, op mix, tie patterns)?
- **Hierarchical planning**: use the trained F_{hi} + macro codes to propose 3-step jumps, refine with F (HWM).
- **Unobserved actions — first results**: variational $q(a \mid s_{t-1}, s_t)$, prior sampling planner, detached intent-decode readout. v1 posterior-collapses; free bits (4 nats) reach 0.195/0.68 — $\approx$22%/70% of the observed-action ceiling. Open frontier: code-taxonomy probes, ranking in code space.
- **Harder worlds**: faithful iGSM (hierarchy + category sums), larger moduli, paraphrase templates, natural-language noise; scale.